# Customer Segmentation Using Distance Geometry and Clustering

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**Abstract**

This project studies the problem of customer segmentation for retail businesses using tools from linear algebra and distance geometry. Customers are represented as vectors in a feature space constructed from transactional data, and similarity is quantified using Euclidean distance. We formulate the clustering task as the minimization of within-cluster variance, which can be written as a sum of squared norms centred at cluster centroids. The K-means algorithm is interpreted as an iterative method that optimises this objective by alternating between assigning points to the nearest centroid and updating centroids as arithmetic means. Using a real-world retail dataset, we demonstrate how this framework produces behaviourally meaningful customer groups that can support targeted marketing and data-driven decision making.

## Contents

# 1 Introduction

Retail businesses interact with large populations of customers whose purchasing behaviour can vary widely over time. Understanding this heterogeneity is essential for designing targeted marketing campaigns, loyalty programmes, and personalised recommendations. Customer segmentation provides a systematic way to partition the customer base into groups that are internally similar and externally distinct.

In this project, customer behaviour is modelled using geometric ideas. Each customer is embedded as a point in a suitable feature space, and the distance between two points reflects how similar or dissimilar their purchasing patterns are. Clustering algorithms then group nearby points together, yielding segments that can be interpreted in business terms.

The focus is on distance-based clustering, in particular the K-means algorithm, which is tightly connected to basic concepts from linear algebra such as vectors, norms, and quadratic forms. This connection allows a mathematically transparent description of the segmentation process and its optimisation objective.

## 1.1 Customer Segmentation in Retail

Customer segmentation aims to divide the overall customer population into relatively homogeneous groups. Typical applications include identifying high-value customers, recognising infrequent or at-risk customers, and tailoring promotions to specific segments. Instead of manually defining such groups, we employ an automatic procedure based on transactional data.

## 1.2 The Role of Distance and Geometry

To apply clustering, each customer must be represented numerically. We construct a feature vector for each customer using recency, frequency, monetary value, and other aggregate transactional statistics. The Euclidean distance between two feature vectors encodes how different their behaviour is in this space. Geometrically, a segment corresponds to a cluster of points that are close to a common centre.

## 1.3 Clustering in Machine Learning

Clustering is an unsupervised learning task: there are no predefined labels, and the goal is to discover underlying structure directly from the data. Among many clustering methods, K-means is particularly appealing because its objective function can be written purely in terms of squared Euclidean distances to cluster centroids, making it a natural setting in which to apply linear algebra and distance geometry.

## 1.4 Project Objective

The objective of this project is to use concepts from linear algebra to formulate and implement a distance-based customer segmentation model. Specifically, we:

- represent customers as vectors in a feature space derived from real transactional data,
- define and analyse a distance-based clustering objective,
- implement the K-means algorithm to minimise within-cluster variance, and
- interpret the resulting segments in terms of purchasing behaviour.

# 2 Mathematical Background

This section summarises the linear algebra and distance geometry concepts used in the project. These ideas provide the framework for representing customers as points and for defining the clustering objective.

## 2.1 Vectors, Norms, and Distance Measures

A column vector in $\mathbb{R}^d$ is an ordered list of $d$ real numbers. In this project, the $d$ components of a vector encode different behavioural features for a single customer, such as recency, frequency, and monetary value:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} \in \mathbb{R}^d.$$

The Euclidean norm of a vector $x \in \mathbb{R}^d$ is defined by

$$\|x\|_2 = \sqrt{x_1^2 + x_2^2 + \cdots + x_d^2}.$$

The Euclidean distance between two customers $x_i$ and $x_j$ is then

$$d(x_i, x_j) = \|x_i - x_j\|_2.$$

This distance quantifies how far apart their feature profiles are in the geometric sense.

## 2.2 Inner Products and Similarity

The standard inner product on $\mathbb{R}^d$ is given by

$$\langle x,\, y\rangle = x^\top y = \sum_{m=1}^{d} x_m y_m.$$

It is related to the norm by $\|x\|_2^2 = \langle x, x\rangle$. Inner products are useful for defining angles and cosine similarity between vectors; however, in this project we primarily use the induced Euclidean norm and distance.

## 2.3 The K-means Objective as a Quadratic Form

Suppose there are $n$ customers with feature vectors $x_1, \dots, x_n \in \mathbb{R}^d$. The K-means algorithm partitions the index set $\{1, \dots, n\}$ into $K$ clusters $C_1, \dots, C_K$ and associates with each cluster $C_k$ a centroid $\mu_k \in \mathbb{R}^d$.

The K-means objective function is the total within-cluster sum of squared distances:

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2.$$

For a fixed cluster $C_k$, the contribution

$$J_k = \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2$$

is a quadratic function of the centroid $\mu_k$. The overall objective is therefore a sum of quadratic terms, and can be viewed as a quadratic form in the centroid variables.

## 2.4 Properties of Euclidean Space Relevant to Clustering

Several basic properties of Euclidean space are important for clustering:

- The squared norm $\|x_i - \mu_k\|_2^2$ is always non-negative.
- For a fixed cluster assignment, the function $J_k$ is strictly convex in $\mu_k$, guaranteeing a unique minimiser.
- The arithmetic mean minimises the sum of squared distances: for fixed $C_k$, the vector

  $$\mu_k^* = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$$

  uniquely minimises $J_k$.

These properties justify the centroid update step in the K-means algorithm.

# 3 Problem Formulation: Distance-Based Customer Segmentation

We now formalise the customer segmentation problem in the language of linear algebra and distance geometry.

## 3.1 Representing Customers as Feature Vectors

Let there be $n$ customers. For each customer $i$, we construct a feature vector

$$x_i = \begin{bmatrix} x_{i1} \\ x_{i2} \\ \vdots \\ x_{id} \end{bmatrix} \in \mathbb{R}^d,$$

where the components can include:

- **Recency:** days since the most recent transaction,
- **Frequency:** number of transactions in a given period,
- **Monetary value:** total amount spent,
- **Quantity:** total items purchased,
- **Average order value:** ratio of total spending to number of invoices.

Standardisation is applied so that all features contribute comparably to Euclidean distances.

## 3.2 Defining the Distance Measure

Distances between customers are computed using the Euclidean metric

$$d(x_i, x_j) = \|x_i - x_j\|_2.$$

Because we work in a standardised feature space, a unit change in any coordinate has a comparable effect on the distance. Customers with similar purchasing behaviour are positioned close together, while dissimilar customers are farther apart.

### 3.3 Within-Cluster Variance and Objective Function

Given a partition of customers into clusters $C_1, \ldots, C_K$ with associated centroids $\mu_1, \ldots, \mu_K$, the within-cluster variance is

$$J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2.$$

The segmentation problem is to find a partition and centroids that minimise $J$. Small values of $J$ correspond to clusters that are geometrically tight around their centroids.

### 3.4 Optimisation Problem and Centroid Updates

The K-means formulation is:

$$\begin{aligned} \underset{\{C_k\},\{\mu_k\}}{\text{minimise}} \quad & J = \sum_{k=1}^{K} \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2 \\ \text{subject to} \quad & C_1, \ldots, C_K \text{ form a partition of } \{1, \ldots, n\}. \end{aligned}$$

For fixed assignments $C_k$, minimising $J$ with respect to $\mu_k$ gives the centroid update. For a single cluster,

$$J_k(\mu_k) = \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2.$$

Differentiating with respect to $\mu_k$,

$$\frac{\partial J_k}{\partial \mu_k} = 2 \sum_{x_i \in C_k} (\mu_k - x_i).$$

Setting the gradient to zero gives

$$\sum_{x_i \in C_k} (\mu_k - x_i) = 0 \quad \Rightarrow \quad |C_k|\, \mu_k = \sum_{x_i \in C_k} x_i \quad \Rightarrow \quad \mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i.$$

Thus the optimal centroid is the arithmetic mean of the points in the cluster.

## 4 Numerical Illustration

To make the abstract formulation concrete, we consider a low-dimensional example with a small number of customers and two features.

### 4.1 Toy Dataset and Features

Suppose there are four customers and two features: frequency (number of purchases) and monetary value (total spending). After standardisation, their feature vectors are

$$x_1 = \begin{bmatrix} 0.20 \\ 0.30 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 0.25 \\ 0.35 \end{bmatrix}, \quad x_3 = \begin{bmatrix} -0.30 \\ -0.40 \end{bmatrix}, \quad x_4 = \begin{bmatrix} -0.35 \\ -0.45 \end{bmatrix}.$$

Intuitively, customers 1 and 2 are similar to each other, as are customers 3 and 4.

### 4.2 Manual Distance Computation and Cluster Assignment

Take two initial centroids $\mu_1^{(0)} = x_1$ and $\mu_2^{(0)} = x_3$. For each customer we compute

$$d\Big(x_i,\, \mu_1^{(0)}\Big) = \|x_i - \mu_1^{(0)}\|_2, \qquad d\Big(x_i,\, \mu_2^{(0)}\Big) = \|x_i - \mu_2^{(0)}\|_2,$$

and assign each customer to the nearer centroid. In this example, customers 1 and 2 join cluster 1 while customers 3 and 4 join cluster 2.

### 4.3 Geometric Interpretation in Two Dimensions

In the two-dimensional feature space, the four points form two clearly separated groups. The K-means procedure is geometrically equivalent to placing a perpendicular bisector between the two centroids that divides the space into two half-planes. Iteratively updating centroids and assignments moves the centres toward the geometric middle of each group until convergence.

## 5 Real-World Segmentation on Retail Data

We now apply this framework to a real transactional dataset from an online retail setting.

### 5.1 Dataset and Preprocessing

The dataset contains invoice-level transaction records, including a unique customer identifier, invoice date, item quantity, unit price, and product information. Preprocessing steps include:

- removing rows with missing customer identifiers,
- filtering out cancelled or negative-quantity transactions,
- computing a total price field as quantity times unit price, and
- converting date fields to appropriate date-time formats.

The cleaned transaction table is then aggregated to one row per customer.

## 5.2 Feature Engineering and Scaling

For each customer we compute:

- **Recency:** number of days since the customer's most recent purchase,
- **Frequency:** count of distinct invoices in the observation window,
- **Monetary:** total spending over the period,
- **Quantity:** total units purchased, and
- **Average Order Value:** ratio of total spending to number of invoices.

Because these quantities can be highly skewed, a logarithmic transform is applied to heavy-tailed features, followed by standardisation so that each feature has mean zero and unit variance.

## 5.3 Implementation of K-means and Choice of $K$

The K-means algorithm is implemented using standard numerical libraries. The main hyperparameter is $K$, the number of clusters. To select a reasonable value, we examine:

- the *elbow method*, which plots within-cluster inertia versus $K$ and looks for a point of diminishing returns, and
- the *silhouette score*, which measures for each point how well it fits its own cluster compared to the nearest alternative cluster.

Based on these diagnostics, a value of $K$ is chosen that balances compactness and interpretability.

The core implementation translates the mathematical model directly into Python:

```
import numpy as np
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score

# Standardise features
X_scaled = StandardScaler().fit_transform(X)

# Fit K-means (quadratic form minimisation)
kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
```

```
labels = kmeans.fit_predict(X_scaled)

# Evaluate
inertia = kmeans.inertia_                   # J (within-cluster variance)
sil     = silhouette_score(X_scaled, labels) # cluster separation
```

## 5.4 Evaluation Metrics and Visualisations

Cluster quality is assessed using:

- **Inertia** (final value of $J$): measures overall cluster compactness,
- **Silhouette score**: measures cluster cohesion and separation,
- **Davies–Bouldin index**: lower values indicate better-separated clusters, and
- **Cluster size distribution**: checks for degenerate tiny clusters.

To visualise the geometric structure, Principal Component Analysis (PCA) is applied to project the standardised feature space onto two dimensions, and customers are plotted coloured by cluster label. Additional plots include bar charts of cluster sizes and boxplots of key features across clusters.

## 5.5 Interpretation of Customer Segments

The resulting segments are interpreted in terms of their average recency, frequency, and monetary value:

- **Cluster 1 – Loyal high-value customers:** low recency, high frequency, high monetary value.
- **Cluster 2 – Moderate regular buyers:** medium frequency and medium spending.
- **Cluster 3 – Infrequent low-spend customers:** low frequency and low monetary value.
- **Cluster 4 – At-risk customers:** historically decent spending but high recency, indicating recent inactivity.

These interpretations connect the geometric compactness of each cluster to actionable business categories.

# 6 Conclusion

This project demonstrates how customer segmentation can be formulated and solved in a geometric framework grounded in linear algebra. By representing each customer as a vector in a standardised feature space and measuring similarity through Euclidean distance, we obtain a clear mathematical model for clustering. The K-means objective function is expressed as a sum of squared distances to cluster centroids, and basic calculus on the resulting quadratic form reveals that the optimal centroid is the arithmetic mean of the assigned points.

Through a low-dimensional numerical illustration and a real-world retail dataset, we observe how this framework naturally produces segments that align with intuitive business categories such as loyal customers, occasional buyers, and at-risk customers. The same mathematical structure extends naturally to higher-dimensional feature spaces and to more advanced clustering techniques that use alternative distance measures, density-based methods, or probabilistic models.

Overall, the project highlights that linear algebra and distance geometry are not only abstract mathematical tools but also provide a rigorous and scalable foundation for practical machine learning tasks that inform real business decisions in customer analytics.

# References


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